Harshit Kumar Sandhu

112, Department of Aerospace Engineering, Indian Institute of Science, Bangalore, Karnataka, India – 560012

Email: harshithk@iisc.ac.in | Google Scholar: Link | ORCID iD: 0009-0003-4816-3503

Research Interests

- Solid Mechanics
- Vibrations and Wave Propagation
- Lattice Dynamics and Mechanical Metamaterials
- Non-Hermitian Systems and Phononic Time Crystals

Work Experience

Indian Institute of Science, Bengaluru May 2026 – Present
Project Associate, LEMS, Department of Aerospace Engineering

Indian Institute of Science, Bengaluru Jan 2024 – May 2024
Teaching Assistant, AE 264 – Vibrations

THE GATE ACADEMY Pvt. Ltd., Bengaluru Sep 2018 – Dec 2019
Subject Matter Expert & Faculty

Education

M.Tech (Research) and Ph.D., Aerospace Engineering *Jan 2020 – June 2027*
Indian Institute of Science, Bengaluru, Karnataka, India
Thesis: "Dynamics of Elastic Time Crystals"
Advisor: Dr. Rajesh Chaunsali
GPA: 8.6/10

B.E., Mechanical Engineering *Aug 2012 – June 2016*
Government Engineering College, Raipur, Chhattisgarh, India
GPA: 7.99/10

Honors and Awards

- **Travel Grant**, *Metamaterials 2025 Conference*, AMOLF, The Netherlands (Sep 2025).
  Awarded **EUR 1080** to support international travel and conference participation.
- **GARP Travel Fellowship**, *Indian Institute of Science (IISc)*, Bengaluru.
  Awarded **INR 2,00,000** for attending the Metamaterials 2025 conference.
- **Student Department Representative**, Aerospace Engineering, *Student Council, IISc* (2024–2026).
- **GATE Fellowship**, Ministry of Education, Government of India (2020–2025).
  Full funding for Ph.D. studies at Indian Institute of Science (IISc), Bangalore.
- **All-India Rank 53**, GATE Engineering Sciences (XE), 2019 (**98.5 percentile**).
- **State Rank 625**, Chhattisgarh Pre-Engineering Test (CG-PET), 2012.

Publications

Journal Articles:

3. **H. K. Sandhu**, & R. Chaunsali, *Temporal Band Topology in Dissipative Phononic Time Crystals* (**Submitted**).

2. **H. K. Sandhu**, A. Umashankar & R. Chaunsali, *Observation of a Temporal Topological Interface State in a Parametric Oscillator* (**Submitted**).
1. **H. K. Sandhu**, S. Dutta, and R. Chaunsali, *Wave propagation in an elastic lattice with nonreciprocal stiffness and engineered damping*, The Journal of the Acoustical Society of America, 159 (2), 978–993 (2026). DOI: 10.1121/10.0042349 **(Special Issue: *Active and Tunable Acoustic Metamaterials*)**

Oral & Poster Presentations

International Conferences:

3. *Observation of a Temporal Topological Interface State in a Parametric Oscillator*, APS Global Physics Summit (March Meeting) 2026, Denver, Colorado, March 2026 (Oral – Virtual).
2. *Wave Propagation in an Elastic Lattice with Nonreciprocal Stiffness and Damping*, Metamaterials 2025, Amsterdam, The Netherlands, September 2025 (Poster).
1. *Wave Propagation in Phononic Time Crystal*, ICMAMS 2024, Ramaiah Institute of Technology, Bengaluru, India, December 2024 (Oral).

Institutional Symposia:

3. *Characterizing a Parametric Oscillator using SINDy*, Mechanical Sciences 3rd Divisional Symposium, IISc Bengaluru, March 2025 (Poster).
2. *Unraveling the Behavior of Waves in Phononic Time Crystal*, Aerospace Research Students' Symposium (AERES 2025), IISc Bengaluru, January 2025 (Oral).
1. *Wave Propagation in Phononic Time Crystal*, Aerospace Research Students' Symposium (AERES 2024), IISc Bengaluru, January 2024 (Oral).

Workshops & Training Attended

- **PhD Course: Complex Wave Phenomena in Metamaterials and Topological Mechanics** (5 CFU), Politecnico di Milano, Italy, August 31–September 4, 2026. (Virtual)
- **EUPROMETA Doctoral School, Metamaterials 2025**, METAMORPHOSE VI, Amsterdam, The Netherlands, September 5–6, 2025.
- **CISM–IISc Workshop on Architectured & Multiscale Materials, Soft Robotics, and Data-Driven Modeling**, jointly organized by CISM (Italy) and IISc (India), July 21–25, 2025.

Teaching Experience

- Delivered a **Guest Lecture** on *"Wave propagation in active elastic lattices"* in the course *Wave Propagation in Designed Materials (AE 351A)*, offered by Prof. Rajesh Chaunsali at IISc Bangalore (December 2023).

Mentoring Experience

All mentoring conducted at LEMS Lab, IISc Bangalore.

- **Pranav M N** (Summer Intern, B.Tech Mechanical Engg., IIT Madras) – Active mechanical oscillators for physical reservoir computing. *June–July 2026.*
- **Jayalakshmi K** (M.Tech Research Student, Aerospace Engg., IISc) – Mechanical neural networks and physical reservoir computing in active systems. *Dec 2025–Present.*
- **Vaishnavi Devgire** (Summer Intern, B.Tech Aerospace Engg. & Applied Mechanics, IIEST Shibpur) – Wave propagation in a 1D chain with a local active site. *May–July 2025.*

- **Avinash Umashankar** (Winter Intern, M.Sc. Physics, SASTRA Deemed University) – Nonlinear vibrations in parametric systems. *Jan–June 2024.*

Leadership and Service Activities

Institutional Service, IISc Bangalore

- Blood donor, Independence Day Blood Donation Camp – Tata Memorial Club & Indian Red Cross Society. *Aug 2024.*

Outreach & Workshop Organization

- Lab Volunteer, IISc Open Day 2026 – demonstrated deployable structures and vibration-based nut loosening.
- Volunteer Organizer, CISM–IISc Workshop. *Jul 2025.*
- Hosted lab tour for Class 12 students, IISc–Agastya Science Enrichment Program. *2024.*
- Lab Volunteer, IISc Open Day 2020 – demonstrated aeroservoelasticity and structural dynamics concepts.

IISc Courses

- **Mathematics and Numerical Methods:** Mathematical Methods for Aerospace Engineers *(AE 211)*, Numerical Solution of Differential Equations *(DS 289)*
- **Solid Mechanics:** Finite Element Methods *(ME 257)*, Vibrations *(AE 264)*.
- **Fluid Mechanics:** Fluid Mechanics *(ME 201)*, Fluid Turbulence *(ME 298)*
- **Advanced Multi-Physics Courses:** Aeroelasticity *(AE 255)*, Fluid–Structure Interaction *(AE 291)*

Technical Skills

- **Programming:** C++, MATLAB, Python
- **Documentation Tools:** LaTeX, Microsoft Office Suite
- **Design and Visualization:** Inkscape, CATIA V5
- **Simulation Software:** COMSOL Multiphysics
- **Experimental Tools:** Laser Doppler Vibrometer (LDV), Data Acquisition Systems (DAQ), 3D Printing

Professional Memberships

- METAMORPHOSE VI Personal Membership: 2025–2026
- Society for Industrial and Applied Mathematics (SIAM): 2021–Present
- American Physical Society (APS): 2023–2024, 2025–2026

Interests

Cooking, singing songs, keyboard playing, reading, yoga, swimming and trekking.